

BEE 4750 Homework 2: Systems Modeling and Simulation

Name:

ID:

Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `c:\Users\holly\4750 HW\hw2-holly`

Installed InvertedIndices	v1.3.1
Installed Libmount_jll	v2.41.1+0
Installed JpegTurbo_jll	v3.1.3+0
Installed InlineStrings	v1.4.5
Installed Accessors	v0.1.42
Installed Preferences	v1.5.0
Installed Roots	v2.2.10
Installed DataFrames	v1.8.0
Installed Fontconfig_jll	v2.17.1+0
Installed WorkerUtilities	v1.6.1
Installed SentinelArrays	v1.4.8
Installed WeakRefStrings	v1.4.2
Installed StatsBase	v0.34.6
Installed DataStructures	v0.19.1
Installed Expat_jll	v2.7.1+0
Installed TableTraits	v1.0.1
Installed PooledArrays	v1.4.3
Installed DataValueInterfaces	v1.0.0
Installed ConstructionBase	v1.6.0
Installed Plots	v1.40.19
Installed PrettyTables	v3.0.8
Installed Ghostscript_jll	v9.55.1+0
Installed CompositionsBase	v0.1.2
Installed Libuuid_jll	v2.41.1+0
Installed InverseFunctions	v0.1.17
Installed CommonSolve	v0.2.4
Installed OpenSSL_jll	v3.5.2+0
Installed Crayons	v4.1.1
Installed Latexify	v0.16.10
Installed Tables	v1.12.1
Installed CSV	v0.10.15
Installed Glib_jll	v2.86.0+0
Installed IteratorInterfaceExtensions	v1.0.0
Installed FilePathsBase	v0.9.24
Installed StringManipulation	v0.4.1
Installed SortingAlgorithms	v1.2.2

Precompiling project...

1340.6 ms	CompositionsBase
1432.8 ms	WorkerUtilities
1392.2 ms	DataValueInterfaces
1622.4 ms	IteratorInterfaceExtensions
1428.4 ms	InverseFunctions

1345.4 ms	InvertedIndices
1410.7 ms	CommonSolve
1335.0 ms	ConstructionBase
1702.6 ms	PooledArrays
1789.6 ms	InlineStrings
1912.6 ms	Crayons
793.5 ms	TableTraits
883.6 ms	LogExpFunctions → LogExpFunctionsInverseFunctionsExt
899.1 ms	InverseFunctions → InverseFunctionsDatesExt
833.0 ms	CompositionsBase → CompositionsBaseInverseFunctionsExt
902.4 ms	ConstructionBase → ConstructionBaseLinearAlgebraExt
1302.0 ms	Preferences
2999.2 ms	DataStructures
1733.9 ms	InverseFunctions → InverseFunctionsTestExt
1804.6 ms	FilePathsBase
1494.1 ms	Unitful → InverseFunctionsUnitfulExt
1303.8 ms	Unitful → ConstructionBaseUnitfulExt
1997.5 ms	SentinelArrays
1394.5 ms	Tables
890.4 ms	PrecompileTools
982.3 ms	JLLWrappers
2088.9 ms	SortingAlgorithms
4539.0 ms	FilePathsBase → FilePathsBaseMmapExt
4544.4 ms	FilePathsBase → FilePathsBaseTestExt
2897.7 ms	RecipesBase
5365.7 ms	Accessors
2886.8 ms	StringManipulation
1040.8 ms	Libmount_jll
1038.1 ms	EpollShim_jll
1221.1 ms	Graphite2_jll
1334.2 ms	OpenSSL_jll
1061.4 ms	Xorg_libICE_jll
1097.0 ms	Xorg_libXau_jll
1124.8 ms	Bzip2_jll
1399.5 ms	LLVMOpenMP_jll
1333.3 ms	libfdk_aac_jll
1349.6 ms	libpng_jll
969.7 ms	LERC_jll
1058.1 ms	LAME_jll
5351.0 ms	ColorSchemes
1398.1 ms	fzf_jll
1538.9 ms	JpegTurbo_jll
1662.0 ms	mtdev_jll

1651.3 ms	Xorg_libXdmcp_jll
1676.6 ms	Ogg_jll
1643.8 ms	x265_jll
1627.8 ms	x264_jll
1690.5 ms	XZ_jll
1566.0 ms	libaom_jll
1490.7 ms	Zstd_jll
1091.3 ms	Expat_jll
964.4 ms	libevdev_jll
1022.5 ms	Xorg_xtrans_jll
1116.7 ms	eudev_jll
1079.9 ms	Opus_jll
1184.3 ms	LZO_jll
1054.9 ms	Libuuid_jll
1177.5 ms	Libffi_jll
1239.2 ms	Libiconv_jll
1204.7 ms	FriBidi_jll
1316.5 ms	Accessors → LinearAlgebraExt
961.9 ms	Xorg_libSM_jll
1428.9 ms	Accessors → TestExt
1219.0 ms	Pixman_jll
1308.6 ms	FreeType2_jll
1592.8 ms	Accessors → UnitfulExt
1017.5 ms	Xorg_libxcb_jll
1941.8 ms	JLFFzf
1973.2 ms	Ghostscript_jll
1077.8 ms	DBus_jll
1001.4 ms	libinput_jll
1402.2 ms	libvorbis_jll
2587.0 ms	OpenSSL
4282.0 ms	Libtiff_jll
3506.2 ms	Wayland_jll
3052.5 ms	Xorg_libX11_jll
3500.9 ms	GettextRuntime_jll
3212.9 ms	Xorg_xcb_util_jll
7155.5 ms	StatsBase
3231.7 ms	Fontconfig_jll
14244.0 ms	Parsers
4727.0 ms	Roots
1110.6 ms	Xorg_libXext_jll
1179.6 ms	Xorg_libXfixes_jll
1205.8 ms	Xorg_libXrender_jll
1169.5 ms	Xorg_libxkbfile_jll

1124.5 ms	Xorg_xcb_util_image_jll
1112.7 ms	Xorg_xcb_util_keysyms_jll
932.0 ms	Xorg_xcb_util_renderutil_jll
1823.1 ms	Glib_jll
1015.9 ms	Xorg_xcb_util_wm_jll
1133.9 ms	Xorg_libXinerama_jll
1113.3 ms	Xorg_libXi_jll
1132.9 ms	Libglvnd_jll
1166.9 ms	InlineStrings → ParsersExt
1030.2 ms	Xorg_libXrandr_jll
6727.4 ms	Latexify
1836.1 ms	Roots → RootsUnitfulExt
1023.5 ms	Xorg_libXcursor_jll
1010.3 ms	Xorg_xkbcomp_jll
1008.8 ms	Xorg_xcb_util_cursor_jll
2326.9 ms	JSON
1375.9 ms	WeakRefStrings
1679.3 ms	Cairo_jll
1384.4 ms	Latexify → SparseArraysExt
963.4 ms	Xorg_xkeyboard_config_jll
2537.4 ms	UnitfulLatexify
1303.3 ms	xkbcommon_jll
2238.3 ms	HarfBuzz_jll
13040.8 ms	PlotUtils
1693.0 ms	Vulkan_Loader_jll
1636.9 ms	libass_jll
1754.1 ms	Pango_jll
885.1 ms	libdecor_jll
2430.3 ms	Qt6Base_jll
979.1 ms	GLFW_jll
4359.3 ms	PlotThemes
3258.8 ms	FFMPEG_jll
5626.2 ms	RecipesPipeline
2363.1 ms	FFMPEG
5437.3 ms	Qt6ShaderTools_jll
4112.4 ms	GR_jll
4394.5 ms	Qt6Declarative_jll
21364.9 ms	HTTP
959.7 ms	Qt6Wayland_jll
17740.6 ms	CSV
31633.9 ms	PrettyTables
5326.6 ms	GR
46629.1 ms	DataFrames

2942.6 ms Latexify → DataFramesExt
71838.4 ms Plots
5305.8 ms Plots → UnitfulExt
137 dependencies successfully precompiled in 140 seconds. 82 already precompiled.

```
Pkg.add("GraphRecipes")  
using Plots  
using LaTeXStrings  
using CSV  
using DataFrames  
using Roots  
using Random  
using Plots  
using GraphRecipes  
using LaTeXStrings  
using Distributions
```

```
Updating registry at `C:\Users\holly\.julia\registries\General.toml`  
Resolving package versions...  
Installed Pango_jll      v1.56.4+0  
Installed SimpleTraits   v0.9.5  
Installed ChainRulesCore v1.26.0  
Installed StaticArrays   v1.9.15  
Installed Interpolations v0.16.2  
Installed PrettyTables   v3.0.10  
Installed Graphs          v1.13.1  
Installed ColorSchemes    v3.31.0  
Updating `C:\Users\holly\4750 HW\hw2-holly\Project.toml`  
[bd48cda9] + GraphRecipes v0.5.14  
Updating `C:\Users\holly\4750 HW\hw2-holly\Manifest.toml`  
[1520ce14] + AbstractTrees v0.4.5  
[79e6a3ab] + Adapt v4.3.0  
[ec485272] + ArnoldiMethod v0.4.0  
[13072b0f] + AxisAlgorithms v1.1.0  
[d360d2e6] + ChainRulesCore v1.26.0  
[35d6a980] ↑ ColorSchemes v3.30.0 v3.31.0  
[3da002f7] ↓ ColorTypes v0.12.1 v0.11.5  
[c3611d14] ↓ ColorVectorSpace v0.11.0 v0.10.0  
[411431e0] + Extents v0.1.6  
[5c1252a2] + GeometryBasics v0.5.10  
[4d00f742] + GeometryTypes v0.8.5  
[bd48cda9] + GraphRecipes v0.5.14
```

```

[86223c79] + Graphs v1.13.1
[d25df0c9] + Inflate v0.1.5
[a98d9a8b] + Interpolations v0.16.2
[c8e1da08] + IterTools v1.10.0
[46757867] + NetworkLayout v0.4.10
[6fe1bfb0] + OffsetArrays v1.17.0
[08abe8d2] ↑ PrettyTables v3.0.8   v3.0.10
[c84ed2f1] + Ratios v0.4.5
[699a6c99] + SimpleTraits v0.9.5
[90137ffa] + StaticArrays v1.9.15
[1e83bf80] + StaticArraysCore v1.4.3
[efce3f68] + WoodburyMatrices v1.0.0
[5ae413db] + EarCut_jll v2.2.4+0
[36c8627f] ↑ Pango_jll v1.56.3+0   v1.56.4+0
[8ba89e20] + Distributed v1.11.0
[1a1011a3] + SharedArrays v1.11.0

```

Info Packages marked with `↑` and `↑` have new versions available. Those with `↑` may be upgradable.
Precompiling project...

```

1766.1 ms   SimpleTraits
1144.5 ms   EarCut_jll
1402.1 ms   Pango_jll
2064.5 ms   ChainRulesCore
1435.4 ms   ChainRulesCore → ChainRulesCoreSparseArraysExt
1157.9 ms   Roots → RootsChainRulesCoreExt
1928.9 ms   LogExpFunctions → LogExpFunctionsChainRulesCoreExt
5193.7 ms   ColorSchemes
8489.7 ms   StaticArrays
2022.5 ms   ConstructionBase → ConstructionBaseStaticArraysExt
2033.3 ms   StaticArrays → StaticArraysStatisticsExt
1955.1 ms   Adapt → AdaptStaticArraysExt
2809.0 ms   StaticArrays → StaticArraysChainRulesCoreExt
2678.7 ms   Accessors → StaticArraysExt
2994.5 ms   ArnoldiMethod
2977.7 ms   GeometryTypes
3031.0 ms   Interpolations
1897.6 ms   Interpolations → InterpolationsUnitfulExt
5441.7 ms   Graphs
13256.9 ms  GeometryBasics
2846.2 ms   NetworkLayout
27690.7 ms  PrettyTables
2454.5 ms   NetworkLayout → NetworkLayoutGraphsExt
11581.6 ms  GraphRecipes
97343.5 ms  Plots → GeometryBasicsExt

```

```
25 dependencies successfully precompiled in 130 seconds. 230 already precompiled.  
2 dependencies precompiled but different versions are currently loaded. Restart julia to a  
Warning: Module Plots with build ID fafbfcfd-f4e7-b885-0edf-d7e192698b7a is missing from th  
This may mean Plots [91a5bcdd-55d7-5caf-9e0b-520d859cae80] does not support precompilation  
@ Base loading.jl:2541
```

```
ArgumentError: Package Distributions not found in current path.  
- Run `import Pkg; Pkg.add("Distributions")` to install the Distributions package.  
ArgumentError: Package Distributions not found in current path.  
  
- Run `import Pkg; Pkg.add("Distributions")` to install the Distributions package.
```

Stacktrace:

```
[1] macro expansion  
    @ .\loading.jl:2296 [inlined]  
  
[2] macro expansion  
    @ .\lock.jl:273 [inlined]  
  
[3] __require(into::Module, mod::Symbol)  
    @ Base .\loading.jl:2271  
  
[4] #invoke_in_world#3  
    @ .\essentials.jl:1089 [inlined]  
  
[5] invoke_in_world  
    @ .\essentials.jl:1086 [inlined]  
  
[6] require(into::Module, mod::Symbol)  
    @ Base .\loading.jl:2260  
  
[7] eval
```



```

@ .\boot.jl:430 [inlined]

[8] include_string(mapexpr::typeof(REPL.softscope), mod::Module, code::String, filename::String) at .\loading.jl:2734

@ Base .\loading.jl:2734

[9] #invokelatest#2

@ .\essentials.jl:1055 [inlined]

[10] invokelatest

@ .\essentials.jl:1052 [inlined]

[11] (::VSCoServer.var"#217#218"{VSCoServer.NotebookRunCellArguments, String})()

@ VSCoServer c:\Users\holly\.vscode\extensions\julialang.language-julia-1.149.2\scripts\notebook\serve_notebook.jl:105

[12] withpath(f::VSCoServer.var"#217#218"{VSCoServer.NotebookRunCellArguments, String}, path::String)

@ VSCoServer c:\Users\holly\.vscode\extensions\julialang.language-julia-1.149.2\scripts\notebook\serve_notebook.jl:105

[13] notebook_runcell_request(conn::VSCoServer.JSONRPC.JSONRPCEndpoint{Base.PipeEndpoint, Base.PipeEndpoint})

@ VSCoServer c:\Users\holly\.vscode\extensions\julialang.language-julia-1.149.2\scripts\notebook\serve_notebook.jl:105

[14] dispatch_msg(x::VSCoServer.JSONRPC.JSONRPCEndpoint{Base.PipeEndpoint, Base.PipeEndpoint})

@ VSCoServer.JSONRPC c:\Users\holly\.vscode\extensions\julialang.language-julia-1.149.2\scripts\notebook\serve_notebook.jl:105

[15] serve_notebook(pipename::String, debugger_pipename::String, outputchannel_logger::Base.Logging.Logger)

@ VSCoServer c:\Users\holly\.vscode\extensions\julialang.language-julia-1.149.2\scripts\notebook\serve_notebook.jl:105

[16] top-level scope

@ c:\Users\holly\.vscode\extensions\julialang.language-julia-1.149.2\scripts\notebook\serve_notebook.jl:105

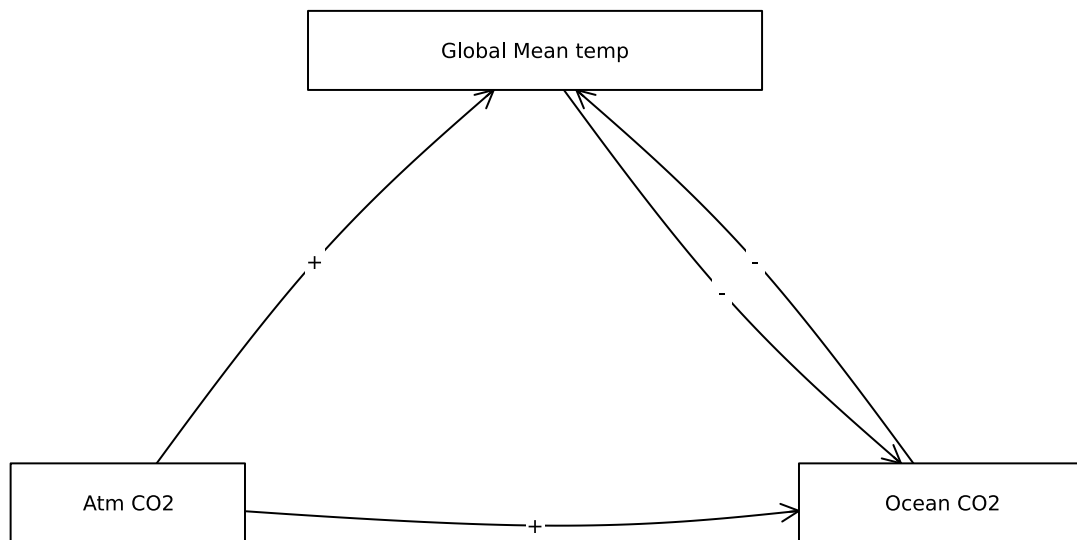
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO₂ concentrations, and ocean CO₂ concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

```
A = [0 0 1 ;  
     1 0 1 ;  
     1 0 0 ]  
  
names = ["Global Mean temp", "Atm CO2", "Ocean CO2"]  
# modify this dictionary to add labels  
edge_labels = Dict((2, 1) => "+", (1,3) => "-", (2,3) => "+", (3,1) => "-")  
shapes=[:rect, :rect, :rect]  
xpos = [0, -0.9, 0.9]  
ypos = [1, 0, 0]  
  
p = graphplot(A, names=names,edgelabel=edge_labels, markersize=0.15, markershapes=shapes, ma  
display(p)
```



From Henry's law, we know that as the partial pressure of CO₂ increases above a body of water, the solubility of CO₂ in the water increases. This led us to choose a "+" for the relationship between Atm CO₂ and Ocean CO₂: As atm CO₂ increases, so will oceanic CO₂. On the other hand, as surface temperature increases, the ability of CO₂ to dissolve in water decreases. So as global mean temperature rises (along with ocean surface temperature), less CO₂ will dissolve, resulting in a "-" relationship. Additionally for this inverse, or dampening, relationship, oceanic carbon is a carbon sink, with more dissolved carbon resulting in slower warming. This solidifies our selection of using "-". We also know from the greenhouse effect that as more CO₂ resides in the atmosphere, more radiation will be trapped in the atmosphere contributing to a rise in temperature. Given this, we chose a "+" relationship.

Tip

Think about Henry's law for CO₂.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays expo-

nentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d .

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

Tip

Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system is in compliance with a regulatory limit of $2.3 \text{ kg/(1000m}^3\text{)}$. You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized “[Snowball Earth](#)”. In this problem, we’ll introduce a simple model of the Earth’s energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C \frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}} + \underbrace{a \ln \left(\frac{[CO_2]}{[CO_2]_{PI}} \right)}_{\text{greenhouse effect}}, \quad (1)$$

where:

- T is the Earth’s global mean temperature (in $^{\circ}\text{C}$);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be $51 \text{ J/m}^2/^{\circ}\text{C}$;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;

- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3 \text{ W/m}^2/\text{°C}$ (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2 \text{ W/m}^2$ (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1 \text{ yr}$.

Problem 3.2

Rather than assuming a constant value for the albedo α , we will represent the ice-albedo feedback by letting α depend on T :

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T + 10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoproterozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C ?

References

List any external references consulted, including classmates.